

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                       | Marks available |          |          |          |          |           |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|-----------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Maths    | Prac     | Total     |
|          |     | (iii) |  | $E = \frac{hc}{\lambda} \quad (1)$<br>$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{564 \times 10^{-9}} = 3.53 \times 10^{-19} \text{ J} \quad (1)$<br>$E = \frac{3.53 \times 10^{-19} \times 6.02 \times 10^{23}}{1000} = 212 \text{ kJ mol}^{-1} \quad (1)$                                                              | 1               | 1        | 1        | 2        |          | 3         |
|          | (d) | (i)   |  | $M_r$ of 2-aminohexanoic acid = 131 (1)<br>moles of 2-aminohexanoic acid = $\frac{0.500}{131} = 3.82 \times 10^{-3} \text{ mol}$<br>gives $3.82 \times 10^{-3} \text{ mol}$ of nitrogen<br>volume of nitrogen = $3.82 \times 10^{-3} \times 24.5 \times 1000 = 93.5 \text{ cm}^3 \quad (1)$<br>ecf possible e.g. from incorrect $M_r$ | 1               | 1        |          | 1        | 1        | 2         |
|          |     | (ii)  |  | award (1) each for any two of following<br>2-aminohexanoic acid is impure<br>temperature <b>less</b> than 298K / pressure <b>greater</b> than 1 atm<br>incomplete reaction<br>nitrogen is lost<br>side reactions                                                                                                                      |                 | 1        | 1        |          | 2        | 2         |
|          |     |       |  | <b>Total question 11</b>                                                                                                                                                                                                                                                                                                              | <b>4</b>        | <b>6</b> | <b>5</b> | <b>3</b> | <b>4</b> | <b>15</b> |